TASK (1) Date: 28/07/2025

conceptual Design USING ER model

Tools required

https://drawio (or (reately / ERDPlus)

Late submission

~~Steps involved in creating ER Diagram.~~

AIM: To design a conceptual design using ER model of college management system.

Steps 1: Understanding

* Analyze the real world application:
college management system.
* understand domain: student, Admission lecuture, subjects.

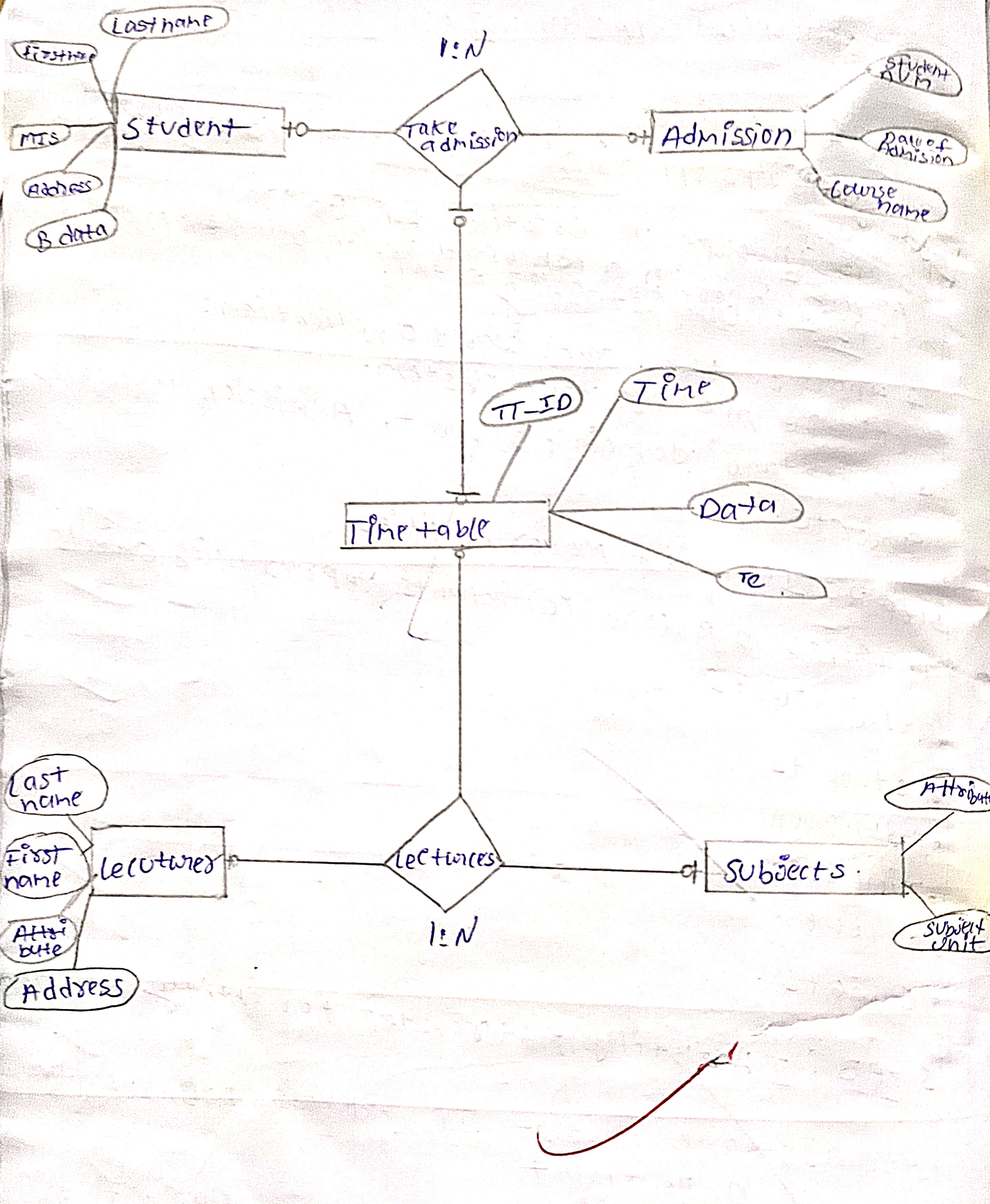
Step 2: Identify major entities

Entities are core components representing objects (or) concepts

- students
- Admission
- Time table
- lecture
- subjects.

Step 3: Indentify Attributes for entity Attribu

- students: Name, student ID, Address, DOB, Department ID, Department
- Admission: Admission NUM, Course name, Date of enrollment, Student ID
- Time table: Time / Data, Classes.



Lastname
firstname
MIS
Address
B data
Student
1:N
Take admission
Admission
Student NUM
Date of Admission
Course name
TT_ID
Time
Timetable
Data
Te
Last name
First name
Attribute
Address
Lecturer
lectures
1:N
Subjects
Attribute
Subject Unit

• Lecturer: Name, Lecturer-ID, Gender, Department, phone number

• Subject: subject-name, subject code

Step 4: Relationship b/w Entities.

* Student take one (or) more Admissions
* Admissions student get time table
* Time table gives one (or) more lecturers.
* lectures teachers one (or) more subjects.

Step 5: DRaw ER Diagram using draw.io

* open https:// draw.io
* choose blank Diagram → click create
* From left panel, drag the following
* use rectangle for entities
* use ellipses for Attributes.
* use dimonds for Relation ships
* connect using lines.
* use labels as (1:N); (M:N) etc to show cardinalities.

Input:

Real time college mangement system.

User Requirements (

Database Design Rules

(entity - Attribute - Relationship identification.

Output:-

Entity relationship diagram (ERD) that shows

* All identified entities with attributes

* All relationship with appropriate cardinatities.

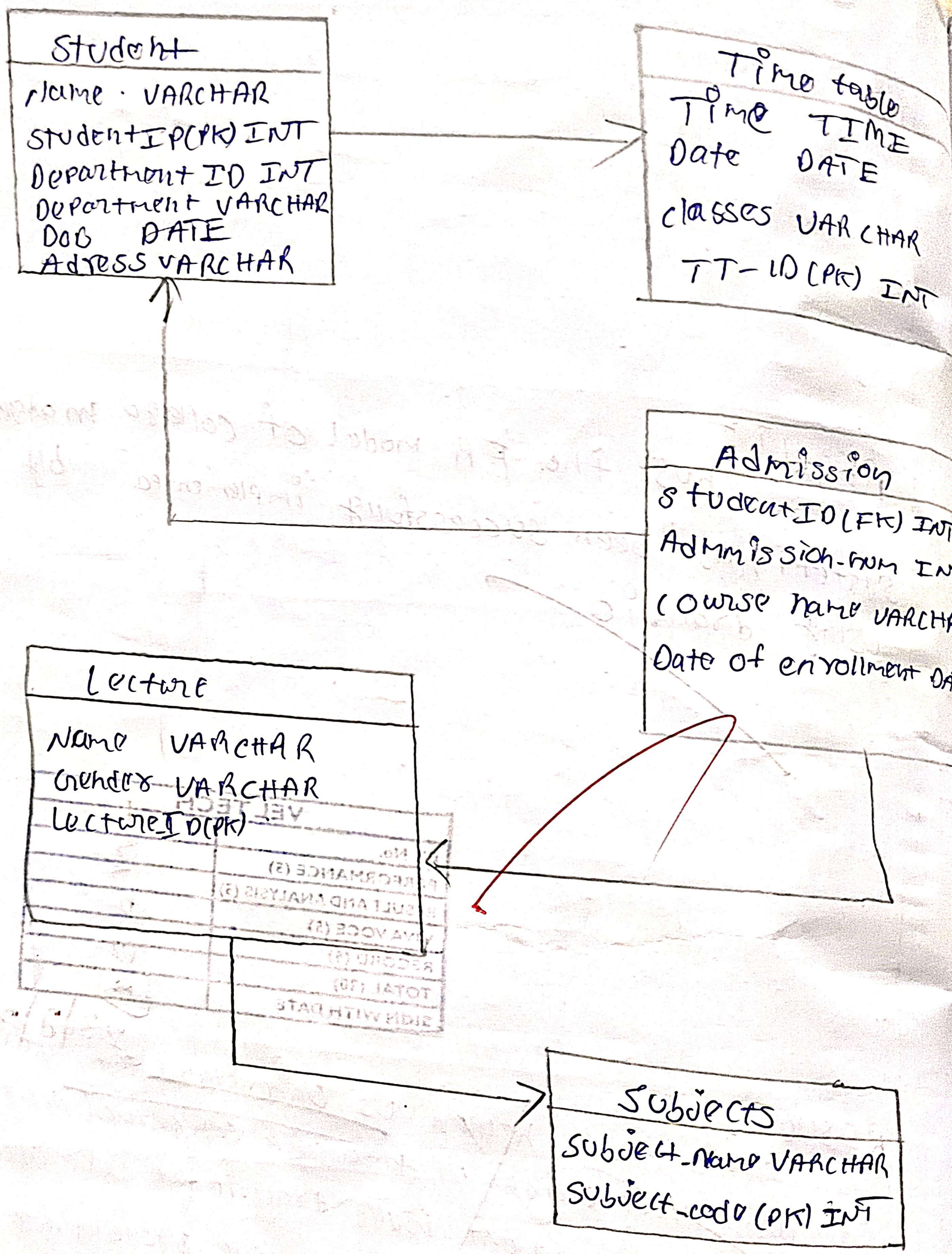
RESULT:-

Thus, the ER model of college mangement system has been successfully implemented by using draw.io

VELTECH	
EX No.	1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	3
VIVA VOCE (5)	2
RECORD (5)	
TOTAL (20)	12
SIGN WITH DATE	28/7/25

RESULT:

This task helped us Understand the importance of conceptual design in database conceptual design in database management. Using draw.io we were able to visually model a real time healthcare system into an ER diagram which forms the foundation for relational schema design in the next phase.


Student
name · VARCHAR
studentID(PK) INT
Department ID INT
Department VARCHAR
DOB DATE
Adress VARCHAR
Time table
Time TIME
Date DATE
classes VARCHAR
TT-ID (PK) INT
Admission
studentID (FK) INT
Admmission-num IN
Course name VARCHA
Date of enrollment DA
Lecture
Name VARCHAR
Gender VARCHAR
LectureID(PK)
Subjects
subject_name VARCHAR
subject_code (PK) INT

28/07/2025 TASK(1.B)

Convert ER Diagram into Relational model

AIM: To convert the ER diagram into realtional model

step for converting ER diagram to the realtional model

* Entity type become a table
* All single - valued attributes become a column for the table
* A key Attribute of the entity type represented by primary key
* The multi valued attribute is represent by a separate table
* composite attribute represent by components
* Derived attributes are not considered in the table.

Using these rules, you can convert the ER diagram to tables & coloun and assign the mapping between the tables.

VELTECH	
EX No.	1.)
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	2
RECORD (5)	—
TOTAL (20)	12
SUBMITH DATE	

28/8/2

RESULT: The relation model for the given ER diagram was successfully converted.